

International Marketing Analytics

Logistic regression across countries

WIRTSCHAFTS UNIVERSITÄT WIEN VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

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WT 2026/27

Learning objectives

- **Read a churn logit:** odds, odds ratios and probabilities, by hand and in Python
- **Choose the model:** binary, discrete-time hazard, multinomial or multilevel logit
- **Compare countries correctly:** response styles and the logit scale problem
- **Interpret and report:** average marginal effects, predictions, interactions
- **Use a churn model:** lift, calibration, campaign profit, risk vs uplift
- **Work with open data:** churn datasets, licences and traps

Session roadmap

1	2	3	4	5	6
Foundations	**Churn and models**	**Countries**	**Reporting**	**Using the model**	**Practice**
Odds, odds ratios, probabilities	Drivers, benchmarks, logit family	Response styles and the logit scale	AMEs, interactions, checklist	Lift, calibration, profit, uplift	Data, quiz, exercise, theses
Slides 8–15	Slides 16–23	Slides 24–30	Slides 31–38	Slides 39–47	Slides 48–58

One rule for today: compare probabilities across countries, not coefficients.

Before we start: where does the contract matter more?

One churn logit per country:

Month-to-month coefficient:
1.19 in Germany,
0.81 in Spain.

What does 1.19 mean?
Log-odds, odds or probability?

Can you compare 1.19 and 0.81?
What else differs between the two countries?

What would you report instead?
Think about what a retention manager wants to know.

Simulated data (telco_churn.csv); separate country logits with the churn formula

Churn in practice: about 1 % a month, many definitions

12.2 % **a year:** O2 Telefónica Germany, postpaid churn (O2 brand), annualised, 2024

0.9 % **a month:** Telefónica España headline churn, FY2024, “best since 2013”

0.93 % **a month:** T-Mobile US postpaid phone churn, FY2025

4.6 % **a month:** US premium streaming (SVOD), weighted average, December 2025

Definitions differ: monthly or annualised, postpaid or all contracts, brand or group. 1.0 % a month is about 11.4 % a year: $1 - (1 - 0.01)^{12}$.

Sources: O2 Telefónica Annual Report 2024; Telefónica FY2024 results; T-Mobile US Q4 2025; Antenna Q1 2026

Retention pays, but check the claim

Claims you will hear

- **“Reducing defections 5 % boosts profits 25 % to 85 %”** (Reichheld & Sasser, HBR 1990): 5 % = five percentage points of churn; improvement assumed free
- **Acquiring a customer costs 5 to 25 times more than retaining one** (Gallo, HBR.org 2014): a range across studies, not a parameter

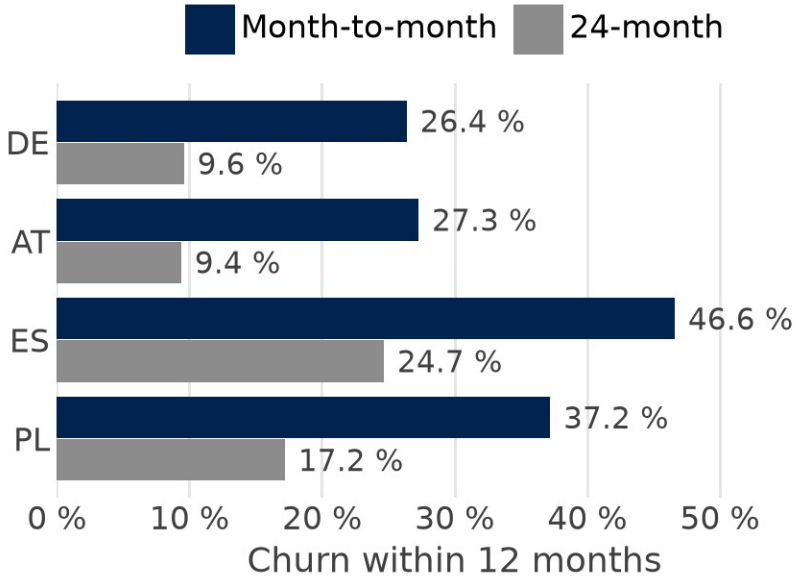
Field experiments

- **Calling likely churners cut churn by 1.9 pp** from 17.2 % (Godinho de Matos et al. 2018)
- **Calling their friends as well:** a further 1.3 pp less churn (same study)
- **Recommending cost-saving plans raised churn:** 10 % vs 6 % (+4 pp) within three months (Ascarza et al. 2016)

Before you quote a number: ask for the outcome, the unit and the counterfactual.

Sources: Reichheld & Sasser 1990; Gallo 2014; Godinho de Matos et al. 2018; Ascarza et al. 2016

Our running example: one operator, four countries



Country	Customers	Churn rate
DE	4,000	15.4 %
AT	1,500	14.6 %
ES	3,000	34.7 %
PL	2,500	28.5 %

Columns: contract type, tenure, monthly fee, price rise, complaint calls, satisfaction 1–5, retention offer (randomised pilot)

Outcome: churn within 12 months

Simulated, with a known truth: the same true effects in all four countries; baselines, customer mix, noise and survey response styles differ.

Simulated data (telco_churn.csv); same true effects in all four countries

Foundations: from odds to probabilities

Part 1 of 6 · why logit, three scales, odds ratios

Why not ordinary regression for churn yes/no?

Linear probability model

- Easy to read: a coefficient is a change in churn probability
- Predicted churn can fall below 0 or above 1
- Same effect at 15 % and at 35 % baseline churn
- Mood (2010): an alternative; in this course: a robustness check

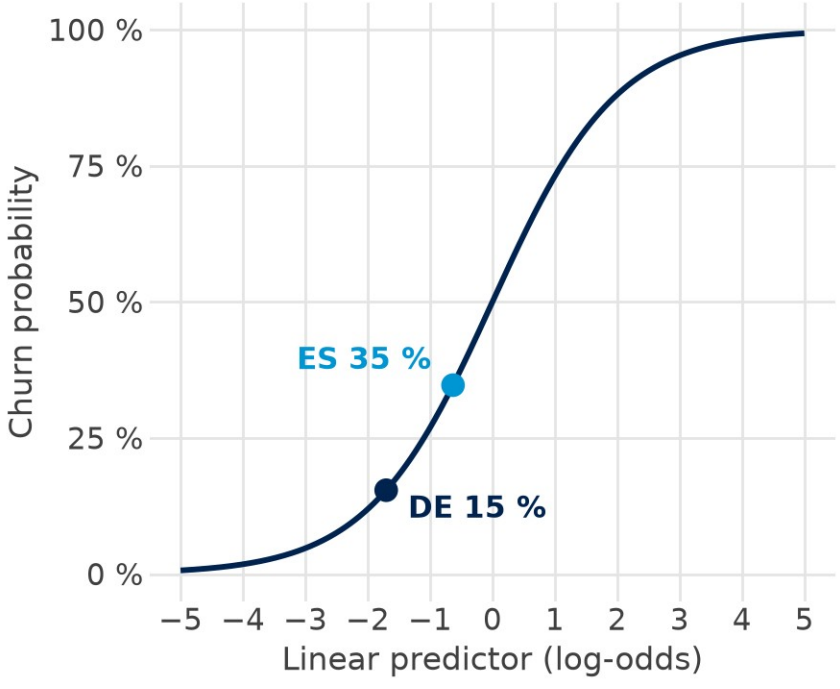
Logistic regression

- Harder to read: a coefficient is a change in log-odds
- Churn probabilities stay between 0 and 1
- Largest effect near 50 %, smallest at the extremes
- Estimated by maximum likelihood

The price of the S-curve: a coefficient no longer tells you how much the probability changes.

Sources: Mood (2010), ESR; Hoetker (2007), SMJ; baselines 15 % and 35 % from simulated data (telco_churn.csv)

One model, three scales



Log-odds: β adds up linearly. This is what the output table shows.

Odds: exp(β) multiplies the odds. This is the odds ratio.

Probability: what a retention manager asks about. Its change depends on the starting point.

Simulated data (telco_churn.csv); curve p = 1 / (1 + exp(−x))

Worked example: German customers, counts to odds

German customers	Churners	Customers	Probability p	Odds = p / (1 − p)
Month-to-month	50	100	0.50	0.50 / 0.50 = 1.0
24-month	20	100	0.20	0.20 / 0.80 = 0.25

Formula

```
p    = churners / customers
odds = p / (1 − p)
     = churners / stayers
```

Odds 1.0: one churner per stayer (50:50)

Odds 0.25: one churner per four stayers

Scales: probability from 0 to 1, odds from 0 to infinity; odds of 1 mean 50:50.

Sources: Invented counts for the calculation (200 German customers)

Odds ratio and log-odds: the same example

4.0 — **Odds ratio:** 1.0 / 0.25
month-to-month: 4 times the odds

−1.39 — **Log-odds, 24-month:** ln 0.25
month-to-month: ln 1.0 = 0

1.39 — **Difference = ln 4.0 = 1.386**
the logit coefficient of month-to-month

Back to probability

```
odds → p:
  p = odds / (1 + odds)
    = 0.25 / 1.25
    = 0.20

log-odds → p:
  p = 1 / (1 + e^1.39)
    = 0.20
```

Not “4 times as likely”: churn rises from 0.20 to 0.50, +30 percentage points (pp) or 2.5 times.

Sources: Invented counts (slide 11)

Your turn: Spanish customers, with a calculator

1 p and odds › 2 Odds ratio › 3 Log-odds › 4 Back to p › 5 Compare

Spanish customers	Churners	Customers	p	Odds	Log-odds
Month-to-month	60	100	?	?	?
24-month	30	100	?	?	?

1–2 probability and odds of both groups; the odds ratio

3–4 log-odds and their difference; from the month-to-month log-odds back to p

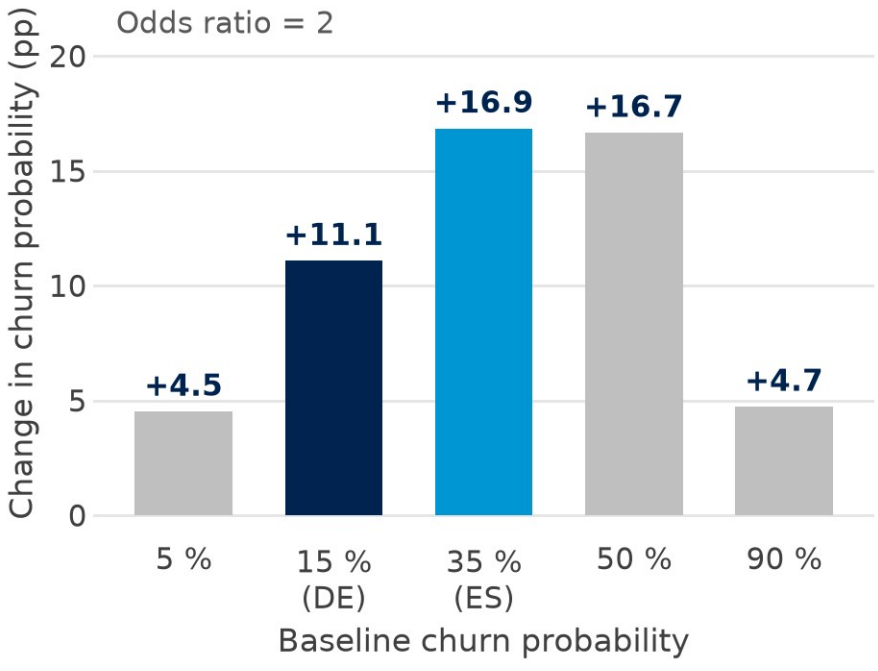
5 compare with Germany: odds ratio and the gap in percentage points

Python check: `smf.logit("churn ~ monthly", data=es).fit().params` logit_churn.qmd, Step 1

Sources: Invented counts for the calculation (200 Spanish customers)

Answers: in class

An odds ratio of 2 is not “twice as likely”



Baseline	With OR 2	Change
5 %	9.5 %	+4.5 pp
15 % (DE)	26.1 %	+11.1 pp
35 % (ES)	51.9 %	+16.9 pp
50 %	66.7 %	+16.7 pp
90 %	94.7 %	+4.7 pp

“Twice as likely” is only a rough approximation for rare outcomes, below about 10 %; annual churn of 15 to 35 % is not rare.

Sources: own calculation, $p_1 = OR \cdot p_0 / (1 - p_0 + OR \cdot p_0)$; DE and ES baselines from simulated data (telco_churn.csv)

Churn as a research context

Part 2 of 6 · drivers, benchmarks, logit family

What drives churn: from theory to variables

Driver	Evidence	Data column
Pricing	Keaveney 1995: one of eight switching causes	price_increase
Service failure	Bolton 1998: failures weigh more than gains	complaints
Satisfaction	Gustafsson et al. 2005: predicts later churn	satisfaction (1–5)
Switching costs	Burnham et al. 2003; Gao et al. 2023: lock-in	monthly (contract type)
Relationship length	Bolton 1998: past satisfaction counts more	tenure_months
Retention action	Ascarza 2018: risk ≠ response	retention_offer (randomised)

Theory first: each predictor maps to a reason for leaving, and the reason tells you which retention action fits.

Sources: Keaveney (1995), JM; Bolton (1998), Mktg Sci; Burnham et al. (2003), JAMS; Gustafsson et al. (2005), JM; Gao et al. (2023), JAMS; Ascarza (2018), JMR

Logit stays a strong benchmark for churn

Tournament

Neslin et al. (2006)

- Logit and trees among the most accurate
- Scored on top-decile lift and Gini
- Models held up on data three months later

Benchmark

Verbeke et al. (2012)

- 11 telecom datasets worldwide
- Many classifiers, logit among them, perform alike
- A few variables suffice

Preparation

Coussement et al. (2017)

- Binning and transforms: AUC up to +14.5 %
- Top-decile lift up to +34 %
- A prepared logit competes with 8 ML methods

When ML wins, it wins by tuning and ensembles; logit keeps coefficients that translate into probability effects.

Sources: Neslin et al. (2006), JMR; Verbeke et al. (2012), EJOR; Coussement et al. (2017), DSS; Lemmens & Croux (2006), JMR; Coussement & Van den Poel (2008), ESWA

Across countries: intercepts and slopes differ

Intercepts: baseline churn

- Churn falls as porting gets dearer and slower across EU markets
- Number portability shifts contract mix and market shares: 218 operators, 52 countries
- Higher loyalty overall in China than in the Netherlands

Slopes: driver effects

- Satisfaction → loyalty depends on culture: 3,393 mobile customers, 8 countries
- Switching costs raise or lower switching, depending on development: 451 effects, 25 countries
- Barriers largely universal in Australia and Thailand

Minimum: country fixed effects; then test country × driver differences on the probability scale (AMEs, Part 4).

Sources: Usero Sánchez & Asimakopoulos (2012), Telecom Policy; Wei & Zhu (2018), POM; Zhang et al. (2014), IBR; Aksoy et al. (2013), J Interact Mktg; Pick & Eisend (2016), JIM; Patterson & Smith (2003), J Retail

The logit family for churn questions

Question	Model	Assumption	Example
Churn within 12 months?	Binary logit	Link is convention	Neslin et al. 2006
When do they churn?	Discrete-time hazard	Shape of the hazard	Bolton 1998
Where do churners go?	Multinomial logit	IIA	Swisscom 2026
Cancel, pause or stay?	Nested logit or MNL	Nests fit the decision	Thesis niche, slide 54
Satisfaction 1–5	Ordered logit	Proportional odds	Survey item, slide 27
Many countries	Multilevel logit	Enough countries	None found, slide 23
Rare events, case-control	Prior-corrected logit	Known population rate	King & Zeng 2001

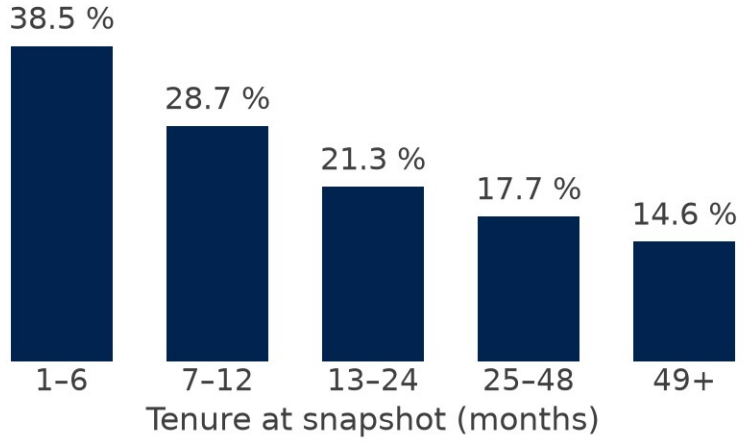
The question decides the model, not the other way round.

Sources: Greene, Chidlow & Strange (2022), IBR; Neslin et al. (2006), JMR; Bolton (1998), Mktg Sci; King & Zeng (2001), Political Analysis; Swisscom Q2 2026 results (Investing.com)

When do customers churn? Discrete-time hazard logit

Customer	Month	Month-to-month	Price letter	Churned
1	1	1	0	0
1	2	1	1	0
1	3	1	1	1
2	1	0	0	0
2	2	0	0	0
2	…	…	…	…

Illustration of the layout, not real data



Simulated data: 12-month churn by tenure

Monthly ↔ annual: annual = $1 - (1 - m)^{12}$; 1.0 % a month ≈ 11.4 % a year

Time-varying predictors: a price-change notice raised monthly churn from 1 % to 1.6 % among Norwegian customers whose costs would fall (Reme et al. 2022)

Sources: Reme et al. (2022), Mgmt Sci; simulated data (telco_churn.csv)

Where do churners go? Multinomial logit and IIA

Destination: multinomial logit

- Outcome: stay, competitor A, competitor B or the own low-cost brand
- Variables describe the customer: one row per customer
- IIA: the odds of A vs B must not change when a new brand enters
- Implausible when two options are close substitutes

A staged decision

1. **Stay or leave?**
2. **If leaving: own low-cost brand or a competitor?**
3. **Which competitor?**

Defend MNL with theory about substitutes, then check with a nested logit.

Sources: Greene, Chidlow & Strange (2022), IBR; Gao et al. (2023), JAMS

Customers nested in countries: how many countries?

4	countries in our data: use country fixed effects, not country random effects
25–30	countries as a rough rule of thumb; with few countries, country-level estimates are unreliable (Bryan & Jenkins 2016)
36	countries: an example of the scale at which culture moderation has been tested (Samaha et al. 2014)
0	multilevel churn studies with customers nested in countries found in our search: a thesis niche

Cross-level question? “Does individualism strengthen the satisfaction effect?” (Frank et al. 2015, six countries) needs many countries and a random slope.

Sources: Bryan & Jenkins (2016), ESR; Samaha et al. (2014), JM; Frank et al. (2015), JEP; literature search, Oct 2026; simulated data (telco_churn.csv)

Comparing countries

Part 3 of 6 · response styles and the logit scale

Two comparability problems stack up

Layer 1: measurement

- Does a satisfaction score of 4 mean the same in Spain and Germany?
- Response styles shift the answers
- Fix: invariance tests, within-country standardisation

20.3 % vs 4.0 %

answer 5: Spain vs Germany

Layer 2: logit scale

- Coefficients are scaled by unexplained variation
- That variation differs by country
- Holds even with perfect measurement

0.81 vs 1.19

coefficient: Spain vs Germany

You need both: invariance checks fix layer 1; probabilities and marginal effects sidestep layer 2.

Sources: Steenkamp & Baumgartner (1998), JCR; Allison (1999), SMR; Mood (2010), ESR; simulated data (telco_churn.csv)

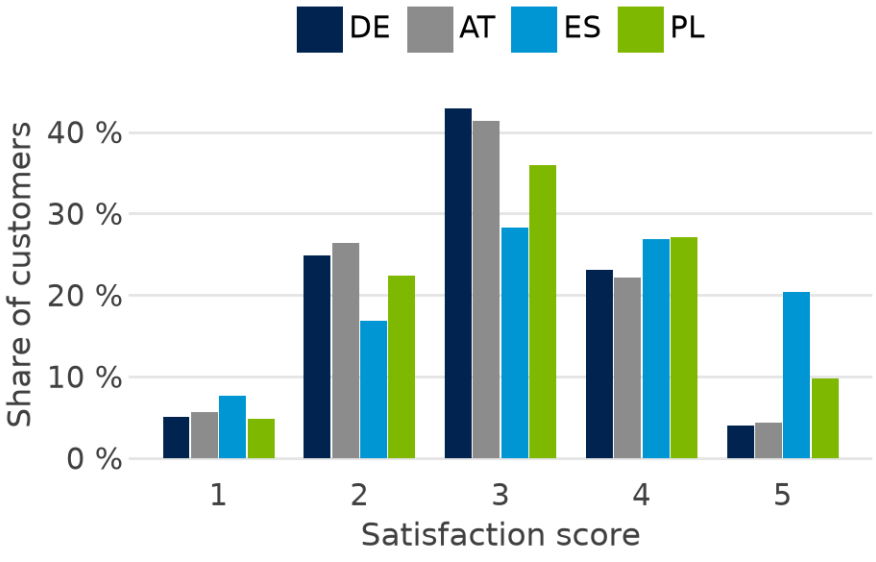
Layer 1: response styles in satisfaction surveys

11	**EU countries:** five response styles bias scale scores and correlations (Baumgartner & Steenkamp 2001)
19	**nations, about 257,000 interviews:** satisfaction levels differ with culture and institutions (Morgeson et al. 2011)
28 %	**of 243 cross-national marketing articles** tested measurement invariance (He, Merz & Alden 2008)
100,040	**car buyers:** the same rated satisfaction goes with different repurchase rates across groups (Mittal & Kamakura 2001)

Configural, metric, scalar: relationships need metric invariance, comparisons of means scalar invariance (Steenkamp & Baumgartner 1998).

Sources: Baumgartner & Steenkamp (2001), JMR; Morgeson et al. (2011), JAMS; He, Merz & Alden (2008); Mittal & Kamakura (2001), JMR

In our data: same satisfaction, different scores



5.1× **top-box answers:** Spain 20.3 %, Germany 4.0 %

3.35 **mean score in Spain** (Germany 2.96)

−0.26 **coefficient per point** in Spain (Germany −0.47)

Equal scores, unequal satisfaction: pool raw 1–5 scores across countries only after a check.

Simulated data (telco_churn.csv); response styles set in the simulation

Layer 2: same effect, different coefficients

Country	True effect	Noise scale	Coefficient	AME (pp)
Germany	1.3	1.0	1.19	15.2
Austria	1.3	1.0	1.02	12.3
Spain	1.3	1.5	0.81	17.3
Poland	1.3	1.2	0.90	16.5

True values set in the simulation; estimates from separate country logits

Latent-variable view

```
coefficient ≈
  true effect / scale

Germany: 1.3 / 1.0 = 1.30
  estimated:         1.19
Spain:   1.3 / 1.5 ≈ 0.87
  estimated:         0.81
```

Compare probabilities:
AMEs and predictions

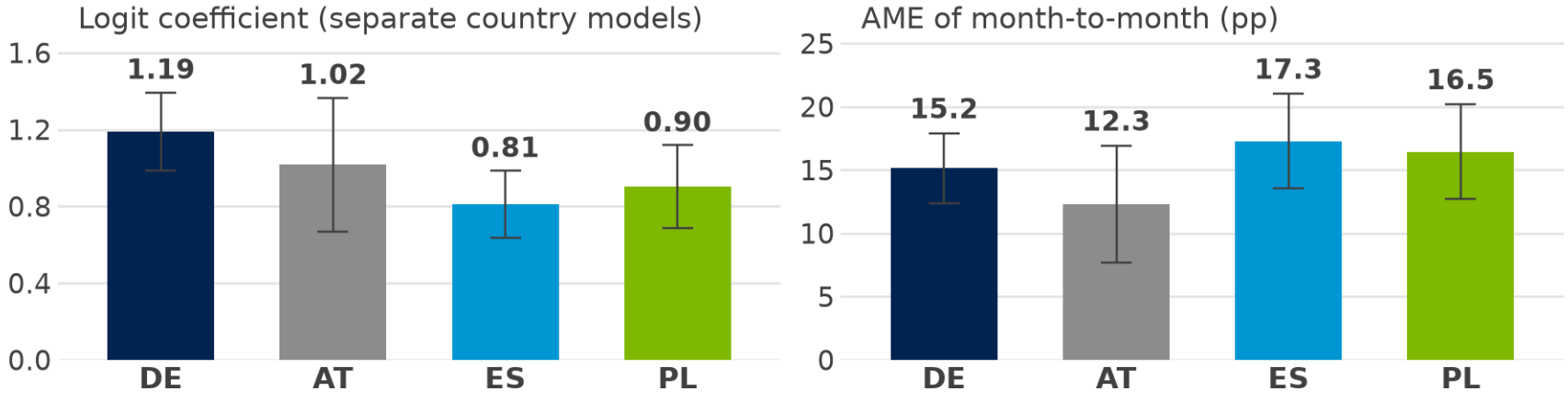
Model the variance:
heterogeneous-choice logit

Nested models:
KHB decomposition

Scale, not strategy: the coefficient is smaller in Spain, the effect on churn is not.

Sources: Allison (1999); Mood (2010); Williams (2009); Karlson et al. (2012); simulated data (telco_churn.csv)

Why the coefficient gap disappears: scale and baseline



Lines: 95 % CIs. AME gap Spain − Germany 2.1 pp (z = 0.90): within sampling error

Scale: more unexplained variation in Spain (1.5 vs 1.0) shrinks its coefficient

Baseline: 34.7 % churn in Spain vs 15.4 % in Germany: nearer 50 %, a steeper S-curve

Sources: Long & Mustillo (2021), SMR; simulated data (telco_churn.csv)

Compare scale-free quantities across countries

AMEs by country

One pooled coefficient: **0.96**

Germany 12.4 pp

Spain 20.1 pp

pooled slope ignores the scale: overstates the gap (separate models: 15.2 vs 17.3 pp)

Predictions

Separate models; month-to-month, price rise, one complaint, 12 months:

Germany 38.1 %

95 % CI 34.1–42.2 %

Spain 58.0 %

95 % CI 54.0–62.0 %

Within a model

β(price rise) / β(contract)

the ratio cancels the scale

like willingness-to-pay ratios in choice experiments

valid only if the model is right

Report on the probability scale: AMEs and predictions with CIs, by country.

Sources: Long & Mustillo (2021), SMR; Swait & Louviere (1993), JMR; simulated data (telco_churn.csv)

Interpreting and reporting

Part 4 of 6 · effects, interactions, checklist

The pooled churn logit: reading the table

Term	Coefficient	Odds ratio	p
Month-to-month	0.962	2.62	< 0.001
Price rise	0.602	1.83	< 0.001
Complaint calls (each)	0.452	1.57	< 0.001
ln tenure (months)	−0.240	0.79	< 0.001
Austria (vs Germany)	−0.010	0.99	0.907
Spain (vs Germany)	1.057	2.88	< 0.001
Poland (vs Germany)	0.618	1.86	< 0.001
Intercept	−2.036	0.13	< 0.001

Complaint calls, OR 1.57: each call raises the odds of churn by 57 %

Tenure, OR 0.85: doubling tenure lowers the odds by 15 %

Fit, pseudo-R^2 0.128: McFadden; fine for a churn model, not a quality score

The table answers "which way, how precise"; "how many pp" needs the next slides.

Simulated data (telco_churn.csv); pooled logit with country dummies

Three quantities to report

Odds ratio

What: multiplies the odds

Use: within one model

Trap: misleads when churn is common

Example: month-to-month OR 2.62

Marginal effect

What: average change in churn probability, in pp

Use: across models and countries

Example: month-to-month +14.2 pp (DE), +17.9 pp (ES), slide 35

Prediction

What: churn probability for a profile

Use: talking to managers

Example:
38.1 % [34.1, 42.2] (DE), 58.0 % (ES), slide 30

Standard: AMEs and predictions with confidence intervals; odds ratios stay in the table.

Sources: Hoetker (2007); Wiersema & Bowen (2009); Zelner (2009); Arel-Bundock (2026); simulated data (telco_churn.csv)

Interactions: the product term is not the effect

Ai & Norton (2003)

- The effect on churn probability varies by customer
- It can be non-zero when the coefficient is zero
- It can even have the opposite sign
- The product-term p-value does not test it

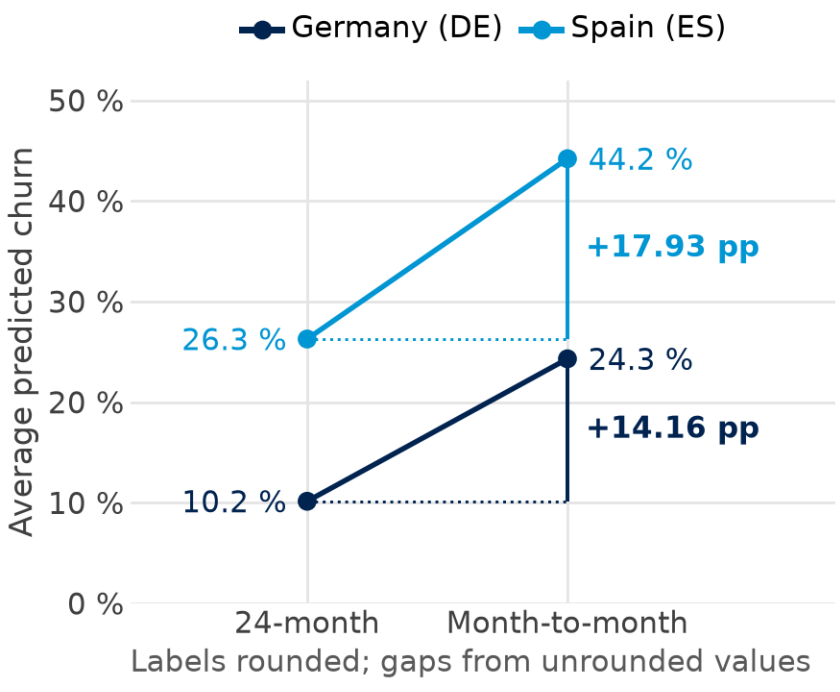
Three steps (Mize 2019)

1. **Plot** predicted churn across the moderator
2. **Test** AMEs at chosen moderator values
3. **Compare** second differences between those AMEs

Managers ask about probabilities: “does the contract matter more in Spain?”

Sources: Ai & Norton (2003), Econ Letters; Mize (2019), Sociological Science

Contract × Spain: −0.245 in log-odds, +3.8 pp



−0.245

Product term: contract × Spain, in log-odds (p = 0.047)

+3.8 pp

Second difference: AME 17.93 pp (ES) minus 14.16 pp (DE); 95 % CI [−0.66, +7.20]

Opposite signs: the product term says "weaker in Spain" (p = 0.047); the probabilities point to "stronger", but the CI includes zero.

Sources: Ai & Norton (2003); Mize (2019); simulated data (telco_churn.csv), interaction model

In Python: marginal effects with statsmodels

Python: statsmodels

```
import numpy as np
import statsmodels.formula.api as smf

m = smf.logit(
    "churn ~ monthly + price_increase"
    " + complaints + np.log(tenure_months)"
    " + C(country, Treatment('DE'))",
    data=df).fit()

m.get_margeff(dummy=True).summary()
ame(m, df, "monthly")    # by country
m.get_prediction(profiles).summary_frame()
```

`get_margeff` average marginal effects in probability units

`ame` by country, pooled slope: 12.4 pp (DE), 20.1 pp (ES); to compare countries see slide 35

`get_prediction` churn probability with 95 % CI, pooled: 35.1 % (DE), 60.9 % (ES)

`plotnine` the curve: predicted churn by tenure and country

Full code and data: `sessions/01-foundations/logit_churn.qmd`, Steps 2, 3 and 7

Simulated data (telco_churn.csv); pooled model of slide 32

Reporting checklist for theses and papers

Element	Report at least	Avoid
Focal effects	AMEs with SEs; predictions for profiles	“x times as likely” for common churn
Moderation	Predictions across the moderator with CIs; second differences	The product-term p-value alone
Countries	AMEs by country; heterogeneous-choice logit	Coefficients or ORs across country models
Survey measures	Invariance tests or within-country standardisation	Pooling raw 1–5 scores across countries
Evaluation	Out-of-sample AUC, top-decile lift, calibration	Accuracy at a 0.5 cut-off
Targeting	Uplift from a randomised control group; campaign profit	Risk scores read as offer effects
Country level	Number of countries; fixed effects below 25–30	Culture effects from 4 countries as proof

Sources: Hoetker (2007); Mize (2019); Long & Mustillo (2021); Williams (2009); Steenkamp & Baumgartner (1998); Bryan & Jenkins (2016); Ascarza (2018)

Endogeneity: price rises and offers are not random

Price rises are chosen

Chosen: firms raise prices where they expect little churn

Notice effect: monthly churn rose from 1 % to 1.6 % after a price-change letter, among customers whose costs would fall (Reme et al. 2022)

Offers are targeted

Targeted: CRM sends offers to the customers with the highest risk

Mixed: an offer coefficient mixes selection and effect; offered customers may churn more because they were picked for their risk

Do: randomise with a control group, as in our pilot (slide 45); otherwise argue an instrument and use a recursive bivariate probit for a binary treatment.

Don't: read the coefficient of a targeted offer or price rise as its effect, or plug first-stage fitted values into a logit.

Sources: Reme et al. (2022), Mgmt Sci; Ascarza (2018), JMR

Using a churn model

Part 5 of 6 · evaluation, targeting, profit, uplift

Train, test, threshold: the confusion matrix

Fit on 70 % → Score 30 % → **Pick a threshold** → Campaign → Measure vs control

Threshold 0.5	Actually churned	Actually stayed
Flagged	175 (TP)	106 (FP)
Not flagged	619 (FN)	2,384 (TN)

3,284 test customers, 794 churners (24.2 %)

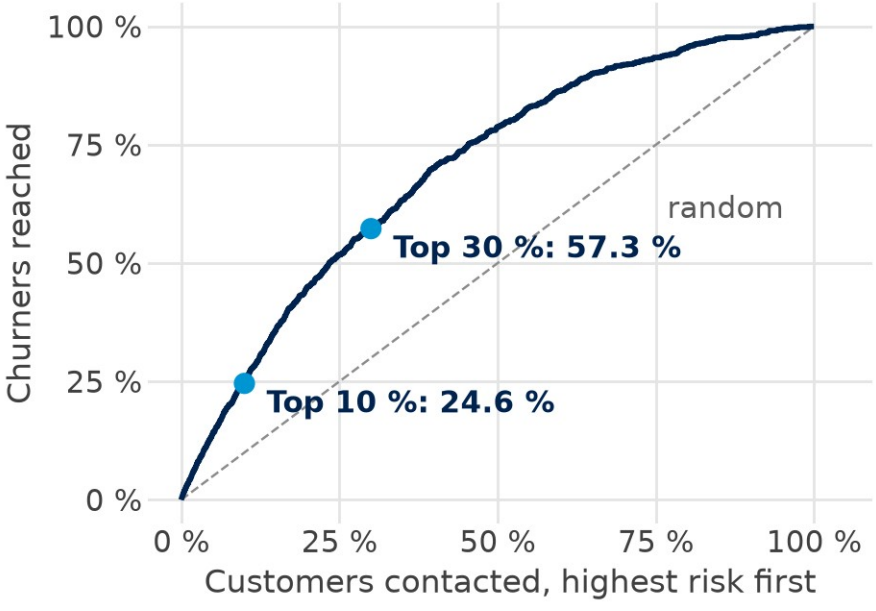
77.9 % **accuracy;** “nobody churns” scores 75.8 %

22.0 % **of churners caught** (recall); 8.6 % of customers flagged

Choose the threshold from costs: flag a customer when p > cost of a wasted offer ÷ (cost of a wasted offer + cost of a missed churner); slide 44’s p* is the same threshold.

Sources: Elkan (2001), IJCAI; simulated data (telco_churn.csv), 3,284 test customers

Ranking quality: AUC, Gini and top-decile lift



0.764 **AUC:** a random churner outranks a random stayer 76 % of the time

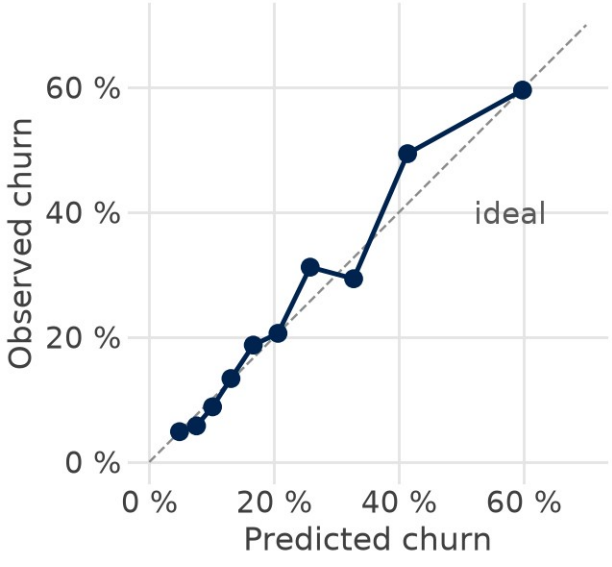
0.529 **Gini** = 2·AUC − 1

2.46 **top-decile lift:** 59.5 % churn in the top 10 % vs 24.2 % overall

Fixed budget? Top-decile lift says what calling 10 % of the base buys.

Sources: Neslin et al. (2006), JMR; simulated data (telco_churn.csv), test customers

Calibration: does 30 % mean 30 %?



0.152 **Brier score;** predicting 24.2 % for everyone scores 0.183

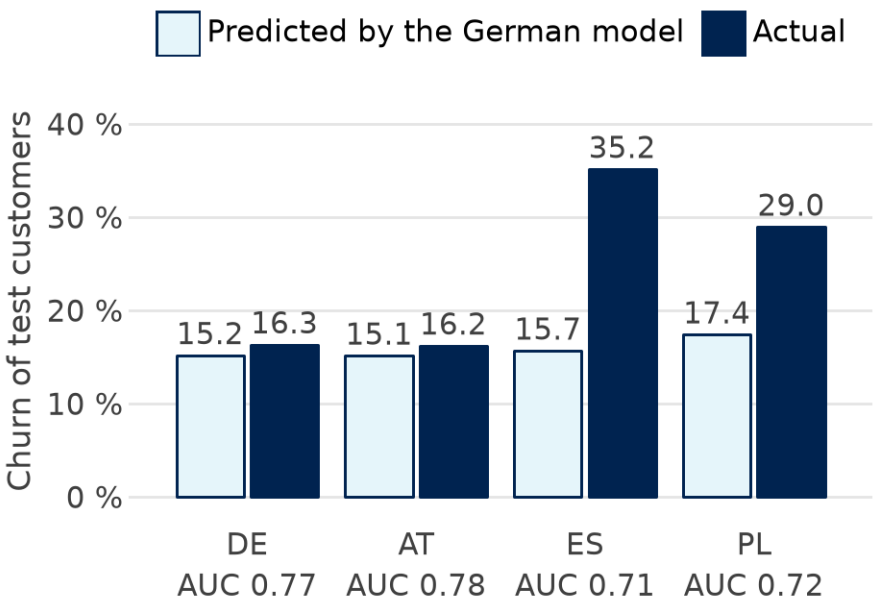
1.06 **calibration slope** (ideal 1); calibration in the large 0.06 (ideal 0)

Do not oversample blindly: 50/50 samples keep the ranking but inflate every probability; correct the intercept

Why it matters: profit rules and CLV use the probability itself, not only the ranking.

Sources: Van Calster et al. (2019), BMC Med; King & Zeng (2001); van den Goorbergh et al. (2022), JAMIA; simulated data (telco_churn.csv)

Does a churn model travel across countries?



0.709 — **AUC of the German model in Spain** (Spain's own model: 0.718)

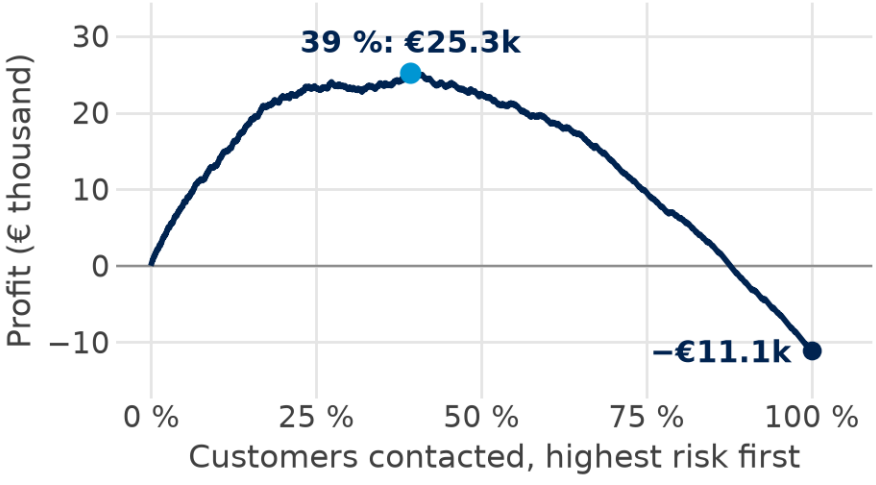
15.7 % — **predicted churn** for Spanish customers; actual 35.2 %

17.4 % — **predicted churn** for Polish customers; actual 29.0 %

Ranking travels, level does not: recalibrate the intercept per country and watch for drift.

Sources: Simester et al. (2020), Mgmt Sci; Risselada et al. (2010), J Interact Mktg; simulated data (telco_churn.csv)

Campaign profit and the cost threshold



0.27 — **break-even churn probability p*:** contact above it

39 % — **of customers contacted** at the profit optimum (€25.3k)

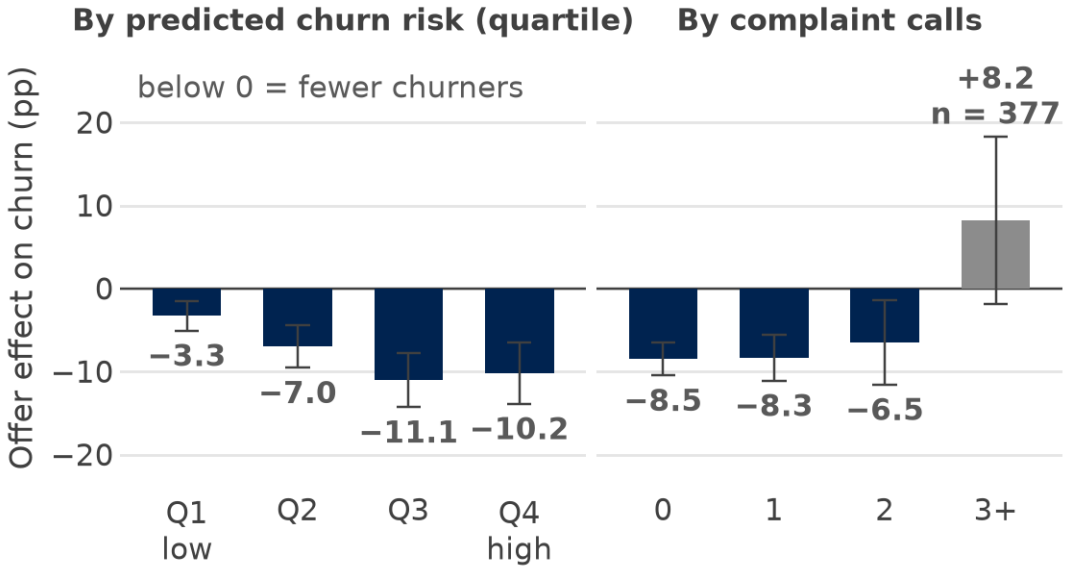
−€11k — **profit** when every customer gets the offer

Neslin et al. (2006) framework (reconstructed); p* derived; A = 0

```
profit = N·α·{[γ·LTV + δ(1 − γ)]·β₀·λ − δ − c} − A
p*     = (c + δ) / (γ·LTV + δ(1 − γ))
illustrative: LTV €400, acceptance γ = 0.25, offer δ = €30, contact c = €3, A = 0
```

Sources: Neslin et al. (2006), JMR; Verbeke et al. (2012), EJOR; Verbraken et al. (2013), IEEE TKDE; simulated data (telco_churn.csv)

Risk is not uplift: the retention-offer pilot



Highest risk ≠ largest effect: Q4 gains no more than Q3 (−10.2 pp vs −11.1 pp)

No response: from three complaint calls on, the offer does not help (44 % churn without it)

Small group: 3+ calls are only 377 customers; the interval includes 0

Target uplift, not risk: it needs a randomised control group.

Sources: Ascarza (2018), JMR; Devriendt et al. (2021), Inf Sci; simulated data (telco_churn.csv), offer pilot

In Python: evaluation and uplift in a few lines

Python: statsmodels + numpy + polars

```
rng = np.random.default_rng(1)
test = rng.random(len(df)) < 0.3
fit = smf.logit(FORMULA + " + satisfaction + C(country)",
                data=df[~test]).fit()
p = fit.predict(df[test]).to_numpy()
y = df.loc[test, "churn"].to_numpy()
top = np.argsort(-p)[:round(0.1 * len(y))]
lift = y[top].mean() / y.mean()          # 2.46

ctrl = df[df["retention_offer"] == 0]
risk = smf.logit(FORMULA + " + satisfaction + C(country)",
                 data=ctrl).fit().predict(df)
df["risk"] = risk
df["risk_group"] = pl.Series(df["risk"]).qcut(
    4, labels=["Q1 low", "Q2", "Q3", "Q4 high"]).to_list()
df.pivot_table("churn", "risk_group", "retention_offer")
```

Split: 70/30 at random, seed 1: 3,284 test customers

Lift: 59.5 % churn in the top decile vs 24.2 %: 2.46

Risk: model fitted on customers without the offer

Uplift: churn with minus without the offer, per risk quartile

Full code: sessions/01-foundations/logit_churn.qmd, Steps 8, 9 and 12.

Simulated data (telco_churn.csv)

Regulation in Europe shapes churn and retention

Rule	What it says	For churn models
EECC, Directive (EU) 2018/1972, Art. 105	Max. 24 months; then one month's notice	Contract ends and notice windows: country-shaped predictors
Germany, § 312k BGB, since 1 July 2022	Cancellation button for online contracts	Cancelling is easy: retention must be proactive
BGH, I ZR 200/25, 16 July 2026	No prominent “pause instead” offer (gym contract; press reports)	No retention offer on the confirmation page: predict earlier
GDPR, Art. 21 and 22	Opt-out from marketing profiling; limits on automated decisions	Honour opt-outs; scores that set prices or terms need care
EU AI Act, Annex III	Credit scoring high-risk; churn models not listed (our reading)	Same logit, different regime: regulation follows the use
USA, FTC click-to-cancel	Rule vacated, Eighth Circuit, 8 July 2025	Exit friction differs by country, so baselines do too

Sources: EUR-Lex 2018/1972; Reg. (EU) 2016/679 (GDPR); § 312k BGB; vzbv and LTO on BGH I ZR 200/25 (July 2026); Reg. (EU) 2024/1689; Latham & Watkins (July 2025). Not legal advice.

Practice: data, quiz and exercise

Part 6 of 6 · open data, quiz, exercise, theses

Open churn datasets for exercises

Dataset	Size, countries	Churn	Licence → use
telco_churn.csv	11,000; DE, AT, ES, PL	12 months, 15–35 %	Simulated, in the repo
IBM Telco Churn	7,043; USA, fictional	Left last month, 26.5 %	Apache-2.0 repo, no data licence: credit IBM
Bank Churn_Modelling	10,000; FR, DE, ES	Closed account, 20.4 %	Synthetic, licence unclear: link to Kaggle
UCI Online Retail II	1.07 m transactions, 40+ countries	You define it (inactivity)	CC BY 4.0: may be shared
Hotel Booking Demand	119,390 bookings, 178 countries	Cancellation, 37.0 %	CC BY 4.0: may be shared
Flash Eurobarometer 243	About 25,000 people, 27 EU countries	Self-reported switching	GESIS: free login, no redistribution

Sources: dataset checks, Oct 2026 (churn_readings_and_data.md); telco_churn.csv is simulated

Open data: traps to check before you fit

1. **Leakage:** drop reservation_status (hotels), Churn Score and Churn Reason (IBM)
2. **Separation:** in Churn_Modelling, 3 or 4 products mean 82.7 % and 100 % churn
3. **Collinearity:** IBM TotalCharges ≈ tenure × MonthlyCharges (r = 0.9996); 11 blanks
4. **Your label, your result:** Online Retail churn depends on the cut-off and the window
5. **Thin country cells:** in Online Retail about 93 % of customers are in the UK

Simulated is fine for learning, but say so: Churn_Modelling looks simulated too.

Sources: dataset checks on the public files, Oct 2026 (churn_readings_and_data.md)

Quiz: odds, countries and targeting

1 A price rise has an odds ratio of 2; baseline churn is 20 %. New churn probability?
A 33 % B 40 % C 22 %

2 Month-to-month coefficient: 1.19 in Germany, 0.81 in Spain. Conclusion?
A larger effect in Germany B none yet C Germany: the coefficient gap is significant (z = 2.8)

3 Month-to-month × Spain: −0.245 (p = 0.047). Effect on churn probability in Spain…
A is smaller B can still be larger than in Germany C is zero

4 You want to know which competitor churners switch to. Model?
A multinomial logit B ordered logit C binary logit

5 Who should get the retention offer?
A the highest predicted risk B those whose churn it lowers most C those with most complaints

3 minutes alone, then compare with your neighbour.

Simulated data (telco_churn.csv)

Answers: in class

Your turn: churn across four countries

Fit · 10′ → **Compare · 10′** → **Evaluate · 10′** → **Target · 5′**

Column	Meaning
country	DE, AT, ES, PL
monthly	1 = month-to-month
satisfaction	survey, 1 to 5
retention_offer	1 = random pilot offer
churn	1 = left in 12 months

1. **Fit** a pooled churn logit with country dummies
2. **Compare** the AME of month-to-month by country with the separate-country coefficients
3. **Evaluate** top-decile lift and a calibration plot on a 30 % test set
4. **Target** the offer effect by complaint calls: who should get it?

Deliverable: one slide per pair with a plot and two sentences for the retention manager.

Simulated data (telco_churn.csv); code in logit_churn.qmd (Steps 2–4, 9, 10, 12 and “Your turn”)

Open niches for theses

1. **Multilevel churn logit:** how much churn sits at country level (15+ countries)?
2. **Model portability:** what do recalibration and reweighting recover in a new country?
3. **Uplift across countries:** are “persuadables” country-specific?
4. **Pause, cancel or stay:** multinomial logit after BGH I ZR 200/25 (July 2026)
5. **Response styles and scale:** model both in one cross-national churn logit

Before you commit: run your own search; these are gaps in our search, not proof of absence.

Sources: Simester et al. (2020); Einav et al. (2025); vzbv and LTO on BGH I ZR 200/25 (Jul 2026)

Research frontiers in churn, 2019–2026

Profit, not risk

Rank by incremental profit (Lemmens & Gupta 2020); learn long-term targeting from surrogates (Yang et al. 2024)

Inertia

Cancellation frictions roughly double subscription revenues (Einav et al. 2025)

Portability

Targeting models lose their edge under covariate and concept shift (Simester et al. 2020)

Fairness

Dropping protected attributes does not remove bias (Ascarza & Israeli 2022)

Text and LLMs

Text from customer contacts adds to structured predictors, it does not replace them (De Caigny et al. 2020)

Networks

Each friend who churns adds 0.06 pp to monthly churn on a 2.15 % baseline (Ferreira et al. 2019)

Sources: Lemmens & Gupta (2020), Mktg Sci; Yang et al. (2024), Mgmt Sci; Einav et al. (2025), AER; Simester et al. (2020), Mgmt Sci; Ascarza & Israeli (2022), PNAS; De Caigny et al. (2020), IJF; Ferreira et al. (2019), JMIS

Key takeaways

1. **Odds are not probabilities:** an odds ratio's meaning depends on the baseline churn rate
2. **The question picks the model:** whether (binary), when (hazard), where to (multinomial)
3. **Two layers:** response styles and the logit scale both threaten country comparisons
4. **Report probabilities:** AMEs and predictions with CIs; interactions as second differences
5. **Use the model for the decision:** lift, calibration, profit; target uplift, not risk

Next: run sessions/01-foundations/logit_churn.qmd before class; it reproduces the model results.

Key sources and further reading

Methods

Ai & Norton (2003). Economics Letters.

Allison (1999). SMR.

Mood (2010). ESR.

Long & Mustillo (2021). SMR.

Mize (2019). Sociological Science.

Hoetker (2007). SMJ.

Swait & Louviere (1993). JMR.

Churn and customers

Neslin et al. (2006). JMR.

Verbeke et al. (2012). EJOR.

Ascarza (2018). JMR.

Lemmens & Gupta (2020). Marketing Science.

Ascarza et al. (2018). Cust. Needs & Solutions.

Aksoy et al. (2013). J Interact Mktg.

Baumgartner & Steenkamp (2001). JMR.

Reading and data guide: `sessions/01-foundations/churn_readings_and_data.md`: articles by strand with DOIs and one-line takeaways, practice figures with source and date, datasets with licences and polars code, thesis niches.

Thank you. Questions and discussion

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